HEART DISEASE PREDICTION USING MACHINE LEARNING

A Data Science Case Study Using Python

Project Domain: Healthcare Analytics | Machine Learning | Predictive Analytics
Dataset: heart.csv
Tools and Technologies: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
Machine Learning Algorithms: Logistic Regression and Random Forest

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1. ABSTRACT

Heart disease is one of the major health concerns worldwide. Early identification of potential heart disease can help healthcare professionals make more informed decisions about further medical evaluation.

This project focuses on analyzing a heart disease dataset using Python and machine learning techniques. The objective is to develop and compare classification models that identify patterns associated with the dataset's target variable.

The project uses the Heart Disease dataset, containing 303 records and 13 input features. The dataset includes information such as age, sex, chest pain type, resting blood pressure, cholesterol levels, maximum heart rate, and other clinical measurements.

Two machine learning algorithms, Logistic Regression and Random Forest, are implemented and evaluated using accuracy, precision, recall, F1-score, and ROC-AUC.

Five-fold cross-validation shows that Logistic Regression achieves a mean accuracy of approximately 84.2%, while Random Forest achieves approximately 82.8%. Random Forest achieves a higher mean ROC-AUC of approximately 0.909.

The findings demonstrate how machine learning can be applied to healthcare-related classification problems. However, the models are educational prototypes and require independent clinical validation before any real-world medical application.

Keywords: Heart Disease Prediction, Machine Learning, Python, Data Science, Logistic Regression, Random Forest, Healthcare Analytics, Predictive Modeling.

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2. INTRODUCTION

Heart disease refers to a group of conditions affecting the heart and its functioning. Identifying individuals who may require further evaluation is an important part of cardiovascular healthcare.

Traditional medical assessment involves clinical examinations, medical history, laboratory tests, and other diagnostic procedures.

Machine learning provides techniques for analyzing datasets, identifying patterns, and developing predictive models based on historical observations.

In this project, machine learning algorithms are applied to a heart disease dataset to investigate whether selected patient characteristics and clinical measurements can help classify the dataset's target variable.

The project follows a structured data science workflow, including data inspection, exploratory data analysis, preprocessing, model development, performance evaluation, and interpretation.

The main objective is to understand how machine learning can be used to explore healthcare data while recognizing the limitations of small datasets and unvalidated predictive models.

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3. PROBLEM STATEMENT

The objective of this project is to develop a machine learning classification system using the features available in the `heart.csv` dataset.

The system aims to identify patterns in the input variables and predict the corresponding target class.

The project addresses the following questions:

1. What characteristics and clinical measurements are available in the dataset?
2. How is the target variable distributed?
3. Can machine learning algorithms distinguish between the two target classes?
4. Which of the selected models performs better across different evaluation metrics?
5. Which features contribute most to the predictions made by the Random Forest model?
6. What limitations must be considered before applying these models to real-world healthcare problems?

Problem type: Supervised learning — binary classification.

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4. OBJECTIVES

The major objectives of this project are:

1. Analyze the structure and characteristics of the heart disease dataset.
2. Perform exploratory data analysis to understand the available features.
3. Examine the distribution of the target variable.
4. Prepare the dataset for machine learning.
5. Implement Logistic Regression as a baseline classification model.
6. Implement Random Forest to explore nonlinear relationships.
7. Evaluate model performance using multiple classification metrics.
8. Compare the models using stratified five-fold cross-validation.
9. Analyze feature importance to understand the fitted Random Forest model.
10. Identify limitations and suggest improvements for future work.

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5. DATASET DESCRIPTION

The project uses a CSV file named `heart.csv`.

The supplied dataset contains 303 observations and 14 columns, consisting of 13 input features and one target variable.

Each row represents one observation, while the columns describe demographic characteristics and clinical measurements.

5.1 Target Variable

The `target` column is the dependent variable that the machine learning models attempt to predict.

It contains two classes:

- **Target = 0:** One outcome category.
- **Target = 1:** The other outcome category.

For the commonly used version of this dataset, these values represent absence and presence of heart disease, respectively. However, the exact coding and meaning should be confirmed using the source documentation before making clinical interpretations.

5.2 Dataset Summary

Characteristic	Value
Total records	303
Input features	13
Target variable	target
Total columns	14
Missing values	0
Target = 0	138
Target = 1	165

The dataset contains approximately 45.5% observations with target = 0 and 54.5% with target = 1.

The target distribution is relatively balanced, although the difference between the classes should still be considered during model evaluation.

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6. TOOLS AND TECHNOLOGIES

The following tools and libraries are used in this project.

Python

Python is the primary programming language used for data analysis and machine learning implementation.

Pandas

Pandas is used to load the CSV file, inspect the dataset, manipulate data, and perform data quality checks.

NumPy

NumPy supports numerical operations and array-based computations.

Matplotlib

Matplotlib is used to create charts and visualize dataset characteristics.

Seaborn

Seaborn can be used to create statistical visualizations, including correlation heatmaps.

Scikit-learn

Scikit-learn provides tools for preprocessing, training classification models, cross-validation, and evaluating model performance.

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7. PROJECT METHODOLOGY

Step 1: Data Collection

Load the supplied `heart.csv` dataset into Python.

Step 2: Data Understanding

Inspect the dataset dimensions, column names, data types, and summary statistics.

Step 3: Data Quality Assessment

Check for missing values, examine the target distribution, and investigate possible duplicate records.

Step 4: Exploratory Data Analysis

Explore relationships between selected features and the target variable using charts and statistical summaries.

Step 5: Data Preparation

Separate the input features from the target variable and divide the dataset into training and testing subsets.

Step 6: Feature Scaling

Standardize the input features within the Logistic Regression pipeline.

Step 7: Model Development

Train Logistic Regression and Random Forest classification models.

Step 8: Model Evaluation

Evaluate the models using accuracy, precision, recall, F1-score, and ROC-AUC.

Step 9: Cross-Validation

Apply stratified five-fold cross-validation to obtain a more robust comparison of model performance.

Step 10: Interpretation: Compare the results, examine Random Forest feature importance, and document the limitations of the analysis.

8. EXPLORATORY DATA ANALYSIS

Exploratory Data Analysis (EDA) is performed to understand the dataset before developing predictive models.

It helps identify patterns, understand feature distributions, and detect potential data quality issues.

8.1 Dataset Structure

The dataset contains 303 observations and 14 columns.

The 13 input features include numerical measurements and integer-encoded categorical variables.

Understanding the data types is important because numerical measurements and categorical codes may require different analytical approaches.

8.2 Missing Value Analysis

A missing-value assessment was performed on the supplied CSV file.

The initial inspection found no missing values.

Therefore, missing-value imputation was not required for this version of the dataset.

However, the absence of missing values does not guarantee that all records are accurate, representative, or free from data quality issues.

8.3 Target Distribution

The target variable contains two classes:

Target Class	Number of Records	Percentage
0	138	45.5%
1	165	54.5%
Total	303	100%

The two classes are relatively balanced.

A bar chart can be used to visualize the number of observations in each class.

This visualization helps determine whether the dataset has a substantial class imbalance that could affect model evaluation.

8.4 Feature Relationships

Relationships between input features and the target variable can be explored using grouped visualizations and statistical summaries.

Examples include:

- Age distribution across target classes.
- Chest pain categories across target classes.
- Maximum heart rate across target classes.
- Exercise-induced angina across target classes.
- Cholesterol and resting blood pressure distributions.
- Correlations between numerical variables.

These analyses can reveal patterns in the supplied data. However, observed associations do not establish causation or prove that a feature independently predicts disease in other populations.

8.5 Correlation Analysis

A correlation heatmap can help identify linear relationships between numerical variables.

Highly correlated features may contain overlapping information.

However, integer-encoded categorical variables require careful interpretation because their numerical codes do not necessarily represent meaningful numerical distances.

Correlation analysis should therefore be treated as an exploratory step rather than definitive evidence of feature importance.

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9. DATA PREPROCESSING

Data preprocessing prepares the dataset for machine learning.

9.1 Separating Features and Target

The dataset is divided into two components:

- **X:** The 13 input features.
- **y:** The target variable.

The target column is excluded from the input features to prevent the model from directly accessing the answer it is expected to predict.

9.2 Train-Test Split

The dataset is divided into training and testing subsets using an 80:20 split.

Subset	Approximate Number of Records
Training data	242
Testing data	61
Total	303

A stratified split is used to approximately preserve the proportion of the two target classes in both subsets.

The random state is set to 42 to make the split reproducible.

9.3 Feature Scaling

Feature scaling is applied within the Logistic Regression pipeline using `StandardScaler`.

Standardization transforms each feature to approximately zero mean and unit variance, based on the training data.

This helps Logistic Regression handle predictors measured on different scales.

Random Forest does not generally require feature scaling, so the original feature values are used for that model.

9.4 Data Leakage Prevention

Preprocessing steps that learn parameters from the data should be fitted using training data only.

Using a scikit-learn pipeline ensures that the scaler is fitted separately within each cross-validation training fold.

This reduces the risk of information leakage during model evaluation.

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10. MACHINE LEARNING MODELS

Two supervised machine learning algorithms are implemented.

10.1 Logistic Regression

Logistic Regression is a classification algorithm commonly used for binary prediction problems.

It estimates the probability that an observation belongs to a particular class.

The predicted probability is converted into a class label using a decision threshold, which defaults to 0.5 in this implementation.

Advantages

- Relatively simple to implement.
- Provides a useful baseline for binary classification.
- Can be easier to interpret than complex ensemble models.
- Supports probability estimation.
- Works effectively when the relationship between predictors and the log-odds is reasonably represented by the model.

Limitations

- May not capture complex nonlinear relationships without additional feature engineering.
- Can be affected by multicollinearity.
- Its performance depends on how well its assumptions represent the data.

In this project, Logistic Regression serves as the baseline model.

10.2 Random Forest Classifier

Random Forest is an ensemble learning algorithm that combines predictions from multiple decision trees.

Each tree is trained using randomized samples and feature selections. The combined predictions can capture nonlinear patterns and interactions.

Advantages

- Can model nonlinear relationships.

- Can capture interactions between features.
- Generally does not require feature scaling.
- Provides feature-importance estimates.
- Can perform well on structured datasets.

Limitations

- May be more difficult to interpret than a simple linear model.
- Feature-importance scores can be misleading when predictors are correlated.
- Can overfit under some configurations.
- Requires careful validation before practical use.

In this project, Random Forest is used to compare an ensemble-based approach against Logistic Regression.

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11. MODEL EVALUATION METRICS

The models are evaluated using five classification metrics.

11.1 Accuracy

Accuracy measures the proportion of predictions that are correct.

Formula:

Accuracy = (True Positives + True Negatives) / Total Predictions

Accuracy provides a general measure of classification performance but may hide important differences between false positives and false negatives.

11.2 Precision

Precision measures the proportion of positive predictions that are actually positive.

Formula:

Precision = True Positives / (True Positives + False Positives)

Higher precision means that a greater proportion of positive predictions are correct.

11.3 Recall

Recall measures the proportion of actual positive cases correctly identified by the model.

Formula:

Recall = True Positives / (True Positives + False Negatives)

Recall is important when missed positive cases are a concern.

However, higher recall can come at the cost of additional false positives.

11.4 F1-Score

The F1-score is the harmonic mean of precision and recall.

Formula:

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

It provides a combined measure of precision and recall.

11.5 ROC-AUC

The Area Under the Receiver Operating Characteristic Curve measures how well a model ranks positive observations above negative observations across different classification thresholds.

A higher ROC-AUC indicates better discrimination between the two classes.

However, ROC-AUC does not measure calibration, establish clinical usefulness, or guarantee good performance on external data.

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12. MODEL TRAINING AND CROSS-VALIDATION

The two classification models are trained using the input features and target variable.

To compare the models more reliably, stratified five-fold cross-validation is performed.

In five-fold cross-validation:

1. The dataset is divided into five approximately equal folds.
2. Four folds are used for training.
3. The remaining fold is used for evaluation.
4. The process is repeated until each fold has served as the evaluation fold.
5. The results are averaged across the five folds.

Stratification helps preserve the target-class proportions within each fold.

The experiment uses a fixed random state for reproducibility.

Cross-validation provides a more informative estimate of model performance than relying on a single train-test split. Nevertheless, results can still vary because the dataset is small.

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13. RESULTS AND MODEL COMPARISON

The following results were obtained using stratified five-fold cross-validation on the supplied dataset.

13.1 Performance Comparison

Evaluation Metric	Logistic Regression	Random Forest
Accuracy	84.2%	82.8%
Precision	82.0%	82.0%
Recall	91.5%	88.5%
F1-Score	86.4%	85.0%
ROC-AUC	0.890	0.909

Values represent mean cross-validation performance, rounded to three decimal places where applicable.

13.2 Interpretation of Results

Logistic Regression

Logistic Regression achieved a mean cross-validation accuracy of approximately 84.2%.

It also achieved a mean recall of approximately 91.5%, indicating that it identified a relatively high proportion of positive target cases across the cross-validation folds.

Its mean ROC-AUC was approximately 0.890.

Random Forest

Random Forest achieved a mean cross-validation accuracy of approximately 82.8%.

Its mean recall was approximately 88.5%, while its mean ROC-AUC was approximately 0.909.

The higher ROC-AUC suggests that Random Forest achieved better average ranking discrimination across thresholds in this experiment, even though its default-threshold accuracy was lower.

Overall Comparison

Logistic Regression performed better in mean accuracy, recall, and F1-score in this experiment.

Random Forest performed better in mean ROC-AUC.

Neither model is universally superior across all evaluation criteria.

The preferred model depends on the intended application, the cost of different errors, and additional validation results.

Important: These results are estimates from a small dataset. They do not establish that either model is suitable for clinical use.

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14. FEATURE IMPORTANCE ANALYSIS

Feature importance is examined using the trained Random Forest model.

Random Forest provides impurity-based importance scores that indicate how much each feature contributes to reducing impurity across the fitted decision trees.

Features with higher importance scores contributed more to the model's decision-making process under this particular importance calculation.

In this dataset, features such as the following may appear among the higher-ranked variables:

- Chest pain category (`cp`)
- Thalassemia-related category (`thal`)
- Maximum heart rate (`thalach`)
- ST depression (`oldpeak`)
- Number of major vessels (`ca`)

The exact ranking depends on the fitted model and its configuration.

A horizontal bar chart can be used to display the most important features.

Interpretation

Feature-importance analysis helps identify which input variables the fitted Random Forest relied on most heavily.

However, these scores must be interpreted cautiously.

They do not prove that a feature causes heart disease or that it is independently predictive in other populations.

Importance scores can also be affected by correlated predictors, feature encoding, and the small sample size.

Further investigation using permutation importance, partial dependence analysis, or SHAP values could provide additional insights.

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15. KEY FINDINGS

The major findings from this project are:

1. The supplied dataset contains 303 records and 13 input features.
2. The initial data inspection found no missing values.
3. The two target classes are relatively balanced.
4. Logistic Regression achieved approximately 84.2% mean cross-validation accuracy.
5. Random Forest achieved approximately 82.8% mean cross-validation accuracy.
6. Logistic Regression achieved higher mean recall and F1-score in this experiment.
7. Random Forest achieved a higher mean ROC-AUC of approximately 0.909.
8. Different evaluation metrics can lead to different conclusions about model performance.
9. Feature importance provides useful exploratory insights but does not establish causation.
10. Independent validation is necessary before considering any real-world application.

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16. LIMITATIONS

Despite the useful results, several limitations must be acknowledged.

16.1 Small Dataset

The dataset contains only 303 observations.

This limits the ability to estimate how well the models will perform on larger and more diverse populations.

16.2 Unknown Representativeness

The dataset may not represent all demographic groups, healthcare settings, or geographical populations.

Results may not generalize to individuals whose characteristics differ from those in the training data.

16.3 Target Definition

The precise meaning and coding of the target variable should be confirmed from the original dataset documentation.

Incorrect assumptions about target labels can lead to misleading conclusions.

16.4 Potential Bias

The project does not provide a comprehensive assessment of subgroup performance or fairness.

The models may perform differently across demographic groups.

16.5 Limited Validation

Five-fold cross-validation provides a useful internal evaluation, but no independent external validation dataset was used.

Repeated or nested cross-validation and external testing could provide stronger evidence of generalizability.

16.6 Threshold Selection

The models use the default classification threshold of 0.5.

This threshold has not been optimized for a specific clinical objective.

16.7 No Clinical Validation

The models have not undergone prospective clinical evaluation or assessment of their effects on patient outcomes.

They should not be used to diagnose heart disease, recommend treatment, or provide reassurance about an individual's health.

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17. FUTURE SCOPE

Several improvements could extend this project.

17.1 Additional Machine Learning Algorithms

Additional models could be evaluated, including:

- Support Vector Machine
- Decision Tree
- K-Nearest Neighbors
- Gradient Boosting
- XGBoost, where available

17.2 Hyperparameter Tuning

Grid Search or Randomized Search could be used to identify promising model configurations.

Any tuning process should be performed within an appropriate cross-validation framework to avoid optimistic performance estimates.

17.3 Improved Feature Interpretation

Permutation importance and SHAP analysis could provide complementary information about how features influence model predictions.

17.4 Model Calibration

Calibration analysis could assess whether predicted probabilities correspond to observed outcome frequencies.

This is particularly important if probabilities are intended to support decisions.

17.5 Independent Validation

Testing the final model on an independent dataset would help assess whether its performance generalizes beyond the supplied sample.

17.6 Subgroup Evaluation

Performance could be examined across relevant demographic groups to identify possible differences in errors and predictive quality.

17.7 Interactive Dashboard

A dashboard developed with Streamlit or Power BI could display dataset summaries, model evaluation metrics, and exploratory visualizations.

Any interactive demonstration should clearly state that the model is educational and not intended for medical decisions.

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18. CONCLUSION

This project demonstrates the application of machine learning techniques to a healthcare-related classification problem using Python.

The analysis covered dataset inspection, exploratory data analysis, preprocessing, model training, cross-validation, performance comparison, and feature-importance interpretation.

Logistic Regression achieved a mean cross-validation accuracy of approximately 84.2%, while Random Forest achieved approximately 82.8%.

Logistic Regression performed better on several default-threshold classification metrics, including recall and F1-score, whereas Random Forest achieved a higher mean ROC-AUC.

These results illustrate why machine learning models should be evaluated using multiple metrics rather than accuracy alone.

The project also highlights the importance of data quality, validation methodology, responsible interpretation, and awareness of dataset limitations.

Overall, the case study provides practical experience in applying Python and machine learning to structured healthcare data.

Further research, independent validation, calibration assessment, and clinical oversight would be necessary before considering any real-world medical application.

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19. TECHNOLOGIES USED

Programming Language: Python

Libraries:

- Pandas
- NumPy
- Matplotlib
- Seaborn
- Scikit-learn

Machine Learning Algorithms:

- Logistic Regression
- Random Forest Classifier

Evaluation Techniques:

- Stratified Train-Test Split
- Stratified Five-Fold Cross-Validation
- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC

Project Type: Supervised Machine Learning — Binary Classification

Dataset: heart.csv

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20. PROJECT DISCLAIMER

This project is intended for educational and portfolio purposes only.

The models were developed using a small dataset and have not been clinically validated.

The reported results should not be interpreted as evidence that the models can reliably diagnose heart disease or predict an individual's medical risk.

The predictions must not be used as a substitute for professional medical advice, diagnosis, or treatment.

Any real-world application would require appropriate data governance, independent validation, clinical evaluation, and regulatory review where applicable.

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21. LINKEDIN PROJECT SUMMARY

Heart Disease Prediction Using Machine Learning | Python Case Study

This project explores the application of machine learning to a heart disease dataset using Python.

The objective was to analyze clinical and demographic features, develop classification models, and compare their performance using multiple evaluation metrics.

The project included data inspection, exploratory data analysis, preprocessing, Logistic Regression, Random Forest, stratified five-fold cross-validation, and feature-importance analysis.

Key results:

- Logistic Regression: 84.2% mean cross-validation accuracy.
- Random Forest: 82.8% mean cross-validation accuracy.
- Random Forest achieved a mean ROC-AUC of 0.909.

Tools used: Python, Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

This project strengthened my understanding of the machine learning workflow, model evaluation, and responsible interpretation of healthcare-related data.